

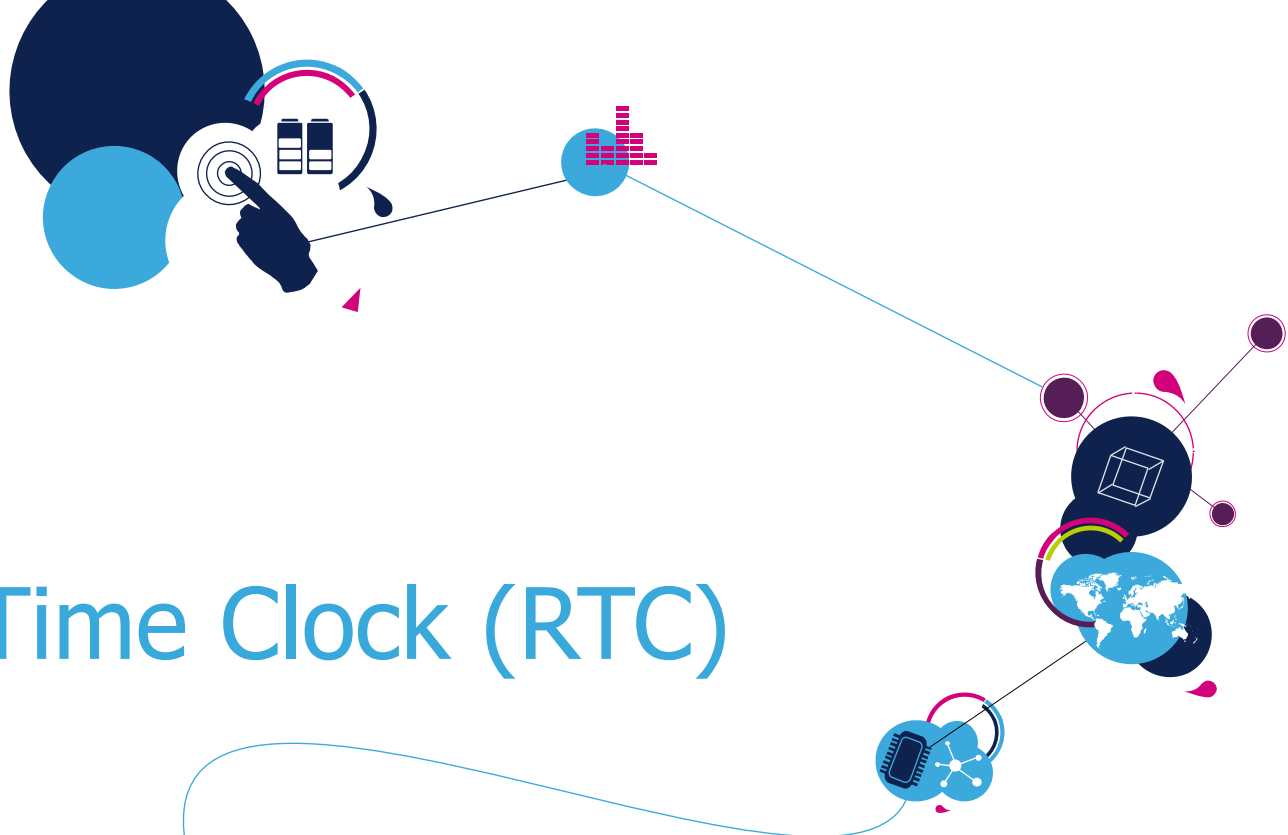


# STM32F3 Technical Training

**For reference only**

**Refer to the latest documents for details**





# Real-Time Clock (RTC)

# RTC Features (1/2)

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- Ultra-low power battery supply current.
- Calendar with Sub seconds, seconds, minutes, hours, week day, date, month, year.
- Daylight saving compensation programmable by software
- Two programmable alarms with interrupt function. The alarms can be triggered by any combination of the calendar fields.
- A periodic flag triggering an automatic wakeup interrupt. This flag is issued by a 16-bit auto-reload timer with programmable resolution. This timer is also called 'wakeup timer'.
- A second clock source (50 or 60Hz) can be used to update the calendar.
- Maskable interrupts/events:
  - Alarm A, Alarm B, Wakeup interrupt, Time-stamp, Tamper detection
- Digital calibration circuit (periodic counter correction) to achieve 0.95 ppm accuracy
- Time-stamp function for event saving with **sub second precision** (1 event)
- Backup registers which are reset when an tamper detection event occurs.
  - 64 bytes for STM32F30x
  - 128 bytes for STM32F37x

# RTC Features (2/2)

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- Alternate function outputs:
  - **RTC\_CALIB**: 512 Hz or 1Hz clock output (with an LSE frequency of 32.768 kHz). It is routed to the device **RTC\_OUT** output.
  - **RTC\_ALARM**: Alarm A, B flag output. It is routed to the device **RTC\_OUT** output.
- Alternate function inputs:
  - **RTC\_TAMP1**: tamper1 event detection.
  - **RTC\_TAMP2**: tamper2 event detection.
  - **RTC\_TAMP3**: tamper3 event detection.
  - **RTC\_TS**: timestamp event detection.
  - **RTC\_REFIN**: reference clock input.
- The RTC clock source could be any of the following three:
  - **LSE oscillator clock**.
  - **LSI oscillator clock**.
  - **HSE divided by 32 in clock controller**.

# RTC overview across families (1/2)

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|                                     | STM32F2x                                                                                     | STM32F0x                                                                                                          | STM32F4x        | STM32F30x | STM32F37x |
|-------------------------------------|----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|-----------------|-----------|-----------|
| RTC in VBAT                         | YES                                                                                          |                                                                                                                   |                 |           |           |
| Calendar in BCD                     | YES                                                                                          |                                                                                                                   |                 |           |           |
| Calendar Sub seconds access         | NO                                                                                           | YES<br>Resolution down to RTCCLK                                                                                  |                 |           |           |
| Calendar synchronization on the fly | NO                                                                                           | YES                                                                                                               |                 |           |           |
| Alarm on calendar                   | 2 wo/ subseconds                                                                             | 1 w/ subseconds                                                                                                   | 2 w/ subseconds |           |           |
| Calendar Calibration                | Calib window :<br><b>64min</b><br>Calibration step:<br>-2ppm/ +4ppm<br>Range [-63ppm+126ppm] | Calib window : <b>8s/16s/32s</b><br>Calibration step: 3.81ppm/1.91ppm/ <b>0.95 ppm</b><br>Range [-480ppm +480ppm] |                 |           |           |

  
life.augmented

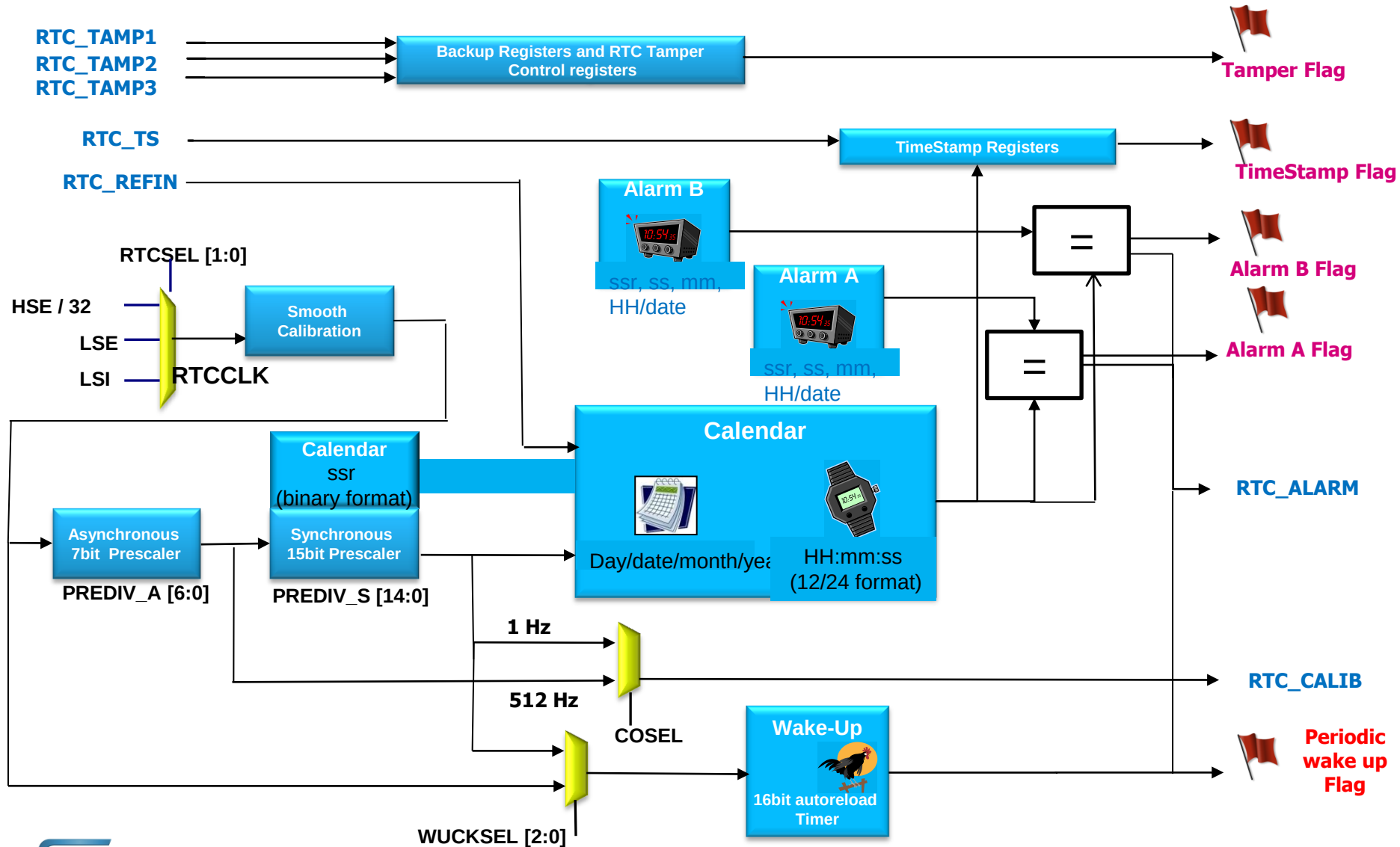
# RTC overview across families (2/2)

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|                                                                  | STM32F2x                                     | STM32F0x                                                           | STM32F4x                                                                  | STM32F30x | STM32F37x |
|------------------------------------------------------------------|----------------------------------------------|--------------------------------------------------------------------|---------------------------------------------------------------------------|-----------|-----------|
| Synchronization on mains                                         | YES                                          |                                                                    |                                                                           |           |           |
| Periodic wakeup                                                  | YES                                          | NO                                                                 | YES                                                                       |           |           |
| Timestamp                                                        | YES<br>Sec, Min, Hour, Date                  | YES<br>Sec, Min, Hour, Date, Sub seconds                           |                                                                           |           |           |
| Tamper                                                           | YES<br>2 pins/1 event<br>Edge detection only | YES<br>2 pins/2 event<br>Level Detection<br>Configurable filtering | YES<br>3 pins/ 3 events<br>Level Detection<br>with Configurable filtering |           |           |
| 32-bit Backup registers                                          | 20                                           | 5                                                                  | 20                                                                        | 16        | 32        |
| PC13-14-15 output state kept in Standby (if not used by RTC/LSE) | NO                                           | YES                                                                | NO                                                                        | YES       |           |

# RTC Block Diagram

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# RTC registers write protection 8

- By default and after reset, the RTC registers are write protected to avoid possible parasitic write accesses.
  - DBP bit must be set in PWR\_CR to enable RTC write access
  - A Key must be written in RTC\_WPR register.
- To unlock write protection on all RTC registers
  - 1. Write '0xCA' into the RTC\_WPR register
  - 2. Write '0x53' into the RTC\_WPR register
    - \* Except for the clear of Alarm and Wakeup timer interrupt flags

→ Writing a wrong key reactivates the write protection.



- The RTC has two clock sources:
  - **RTCCLK** used for RTC timer/counter, can be either the HSE/32, LSE or LSI clocks.
  - **PCLK1** used for RTC register read/write access.
- Before to start using the RTC you have to program the clock controller :
  - Configure and Enable the **RTCCLK** source in the RCC\_BDCR register

# RTC in Low Power Modes and in Reset

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- The RTC remains active what ever the low power mode
  - Sleep, STOP, STANDBY
- When enabled, 5 events can exit the device from low power modes:
  - Alarm A
  - Alarm B
  - Wakeup
  - Tamper 1/ 2 / 3
  - TimeStamp
- The RTC remains active in VBAT mode (VDD off) when clocked by LSE
- The RTC remains active under Reset except at Power-on Reset
  - The RTC configuration registers including prescaler programming are not affected by system Reset else than Power-on Reset.
  - When clocked by LSE, the RTC clock is not stopped under Reset, except power-on reset.

# RTC Alternate function configuration (1/2)

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**RTC pin (PC13) :**

| Pin configuration and function      | RTC_ALARM enabled | RTC_CALIB enabled | Tamper enabled | Time stamp enabled | PC13MODE   | PC13VALUE              |
|-------------------------------------|-------------------|-------------------|----------------|--------------------|------------|------------------------|
| Alarm out output OD                 | 1                 | Don't care        | Don't care     | Don't care         | Don't care | 0                      |
| Alarm out output PP                 | 1                 | Don't care        | Don't care     | Don't care         | Don't care | 1                      |
| Calibration out output PP           | 0                 | 1                 | Don't care     | Don't care         | Don't care | Don't care             |
| TAMPER input floating               | 0                 | 0                 | 1              | 0                  | Don't care | Don't care             |
| TIMESTAMP and TAMPER input floating | 0                 | 0                 | 1              | 1                  | Don't care | Don't care             |
| TIMESTAMP input floating            | 0                 | 0                 | 0              | 1                  | Don't care | Don't care             |
| Output PP forced                    | 0                 | 0                 | 0              | 0                  | 1          | PC13 output data value |
| Standard GPIO                       | 0                 | 0                 | 0              | 0                  | 0          | Don't care             |

OD: open drain; PP: push-pull.

PC13 is available in VBAT mode

# RTC Alternate function configuration (2/2)

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## LSE pin PC14 configuration (1) :

| Pin configuration and function | LSEON | LSEBYP     | PC14MODE   | PC14VALUE              |
|--------------------------------|-------|------------|------------|------------------------|
| LSE oscillator                 | 1     | 0          | Don't care | Don't care             |
| LSE BYPASS                     | 1     | 1          | Don't care | Don't care             |
| Output PP forced               | 0     | Don't care | 1          | PC14 output data value |
| Standard GPIO                  | 0     | Don't care | 0          | Don't care             |

1. OD: open drain; PP: push-pull.

## LSE pin PC15 configuration (1) :

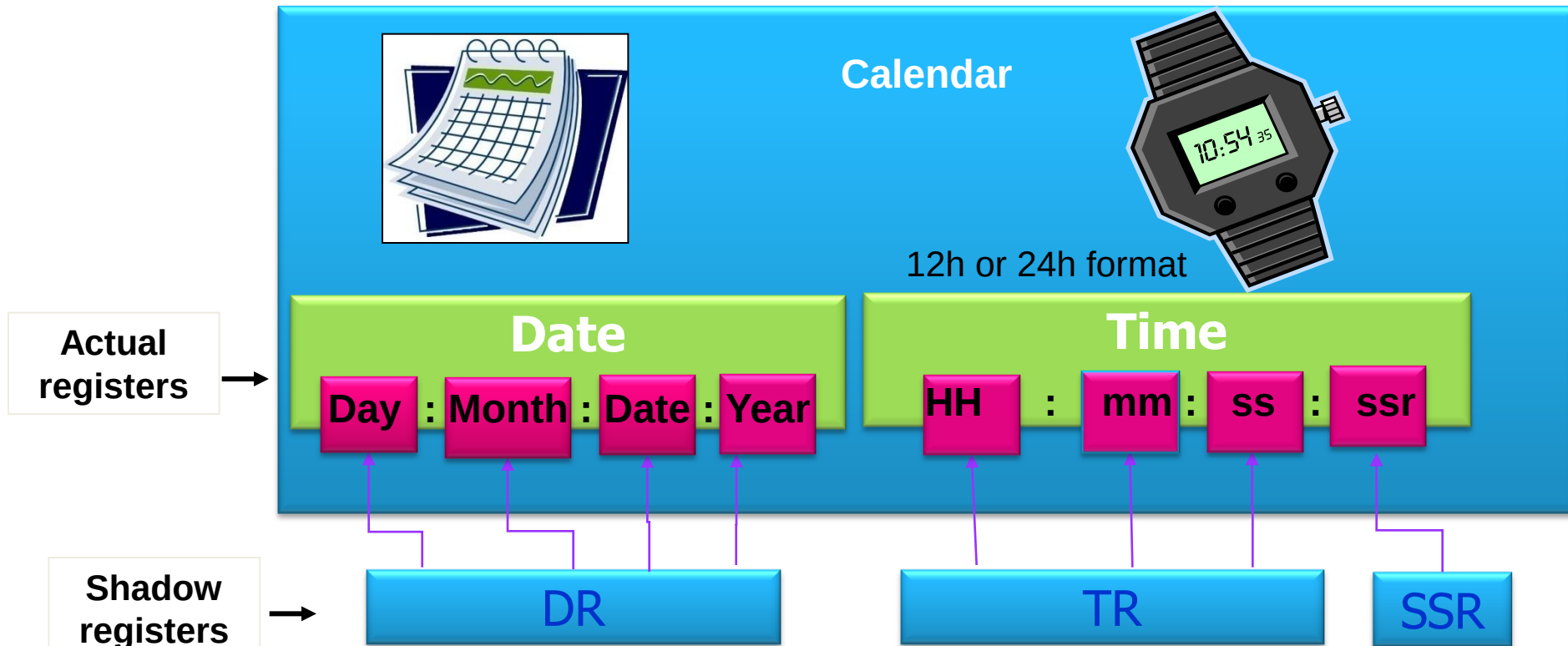
| Pin configuration and function | LSE ON | LSEBYP     | PC15MODE   | PC15VALUE              |
|--------------------------------|--------|------------|------------|------------------------|
| LSE oscillator                 | 1      | 0          | Don't care | Don't care             |
| Output PP forced               | 1      | 1          | 1          | PC15 output data value |
|                                | 0      | Don't care |            |                        |
| Standard GPIO                  | 0      | Don't care | 0          | Don't care             |

1. OD: open drain; PP: push-pull.

# RTC Calendar (1/4)

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- The initialization or the reading of the calendar value is done through 3 shadow registers, SSR, TR and DR. The RTC TR and DR registers are in BCD format.
- SSR register represents the RTC Sub seconds register



- RTC initialization :
  - **Enter in initialization phase mode by setting the INIT bit in ISR register**
    - This mode is confirmed with the INITF flag also in ISR register
  - Program the prescaler register (PRER) according to the clock source to get 1Hz clock to the calendar.
  - **Load the initial date values in the 2 shadow registers (TR, DR).**
    - And other configuration registers like RTC\_CR (hour format, ...)
  - **Exit the initialization phase clearing INIT bit.**
    - The actual calendar register are then automatically loaded and the counting restarts after few RTCCLK clock periods.
- After reset the check of the INITS flag in ISR register indicates if the calendar is already initialized (year not at zero) or not (like after Power-on).
- To manage the “daylight saving” there are 3 bits in CR:
  - SUB1H or ADD1H to subtract or add one hour to the calendar
  - BCK to memorize above action

# RTC calendar (3/4) 15

- The shadow registers are automatically updated each time the RTCCLK clock is synchronized with System Clock.
- The calendar read can be done in 2 different modes :
  - **BYPSHAD=0 : Read shadow registers**
    - RSF flag in ISR register is used to ensure that the calendar value from shadow register is the up-to-date one.
    - Update of DR is frozen **after reading TR** , and unfrozen when **DR is read**.
    - Update of TR and DR is frozen **after reading SSR** , and unfrozen when **DR is read**.
  - **BYPSHAD=1 : Bypass shadow registers**
    - Reading calendar makes direct access to the calendar counters
    - Software must read all calendar registers twice and compare the results to ensure that the data are coherent and correct.

# RTC calendar (4/4) 16

- Calendar can be synchronized up to 1s on the fly by adding/subtracting an offset with the sub second resolution.
  - Allow synchronization to remote clock
- Reference Clock detection: A more precise second source clock (like mains 50 or 60 Hz) can be used to enhance the long-term precision of the calendar:
  - The second source clock is automatically detected and used to update the calendar
  - The LSE clock is automatically used to update the calendar whenever the second source clock becomes unavailable
- Timestamp : Calendar value (including sub-seconds) is saved in Timestamp registers on external I/O event

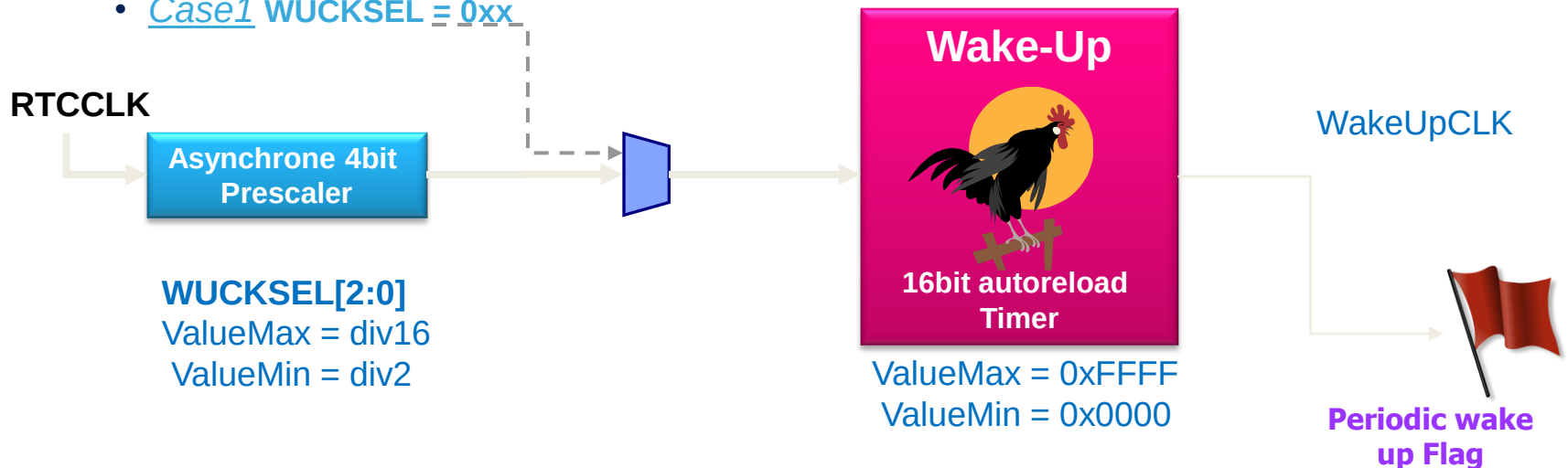
# RTC Programmable Alarm 17

- 2 Full programmable Alarms
- Able to **exit** the device from **STOP/STANDBY modes**.
- **Alarms event** can also be routed to the **specific output pin RTC\_OUT** with configurable polarity.
- The **Alarm flags** are set if the calendar sub seconds, seconds, minutes, hours or date match the value programmed in the alarm registers (ALRMASR & ALRMAR, ALRMBSSR & ALRMBR).
- Calendar sub second, seconds, minutes, hours or date fields can be independently selected (maskable or not maskable).

# WakeUp configuration (1/3)

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- The periodic wakeup flag is generated by a 16-bit programmable binary auto-reload down counter (WUTR registers)
- Able to **exit** the device from **STANDBY modes**.
- The wakeup clock source selection is done via WUCKSEL [2:0] bits in control register RTC\_CR (to program these bits the auto wakeup must be deactivated, WUTE=0).
- **3 possible cases are possible:**
  - Case1 WUCKSEL = 0xx



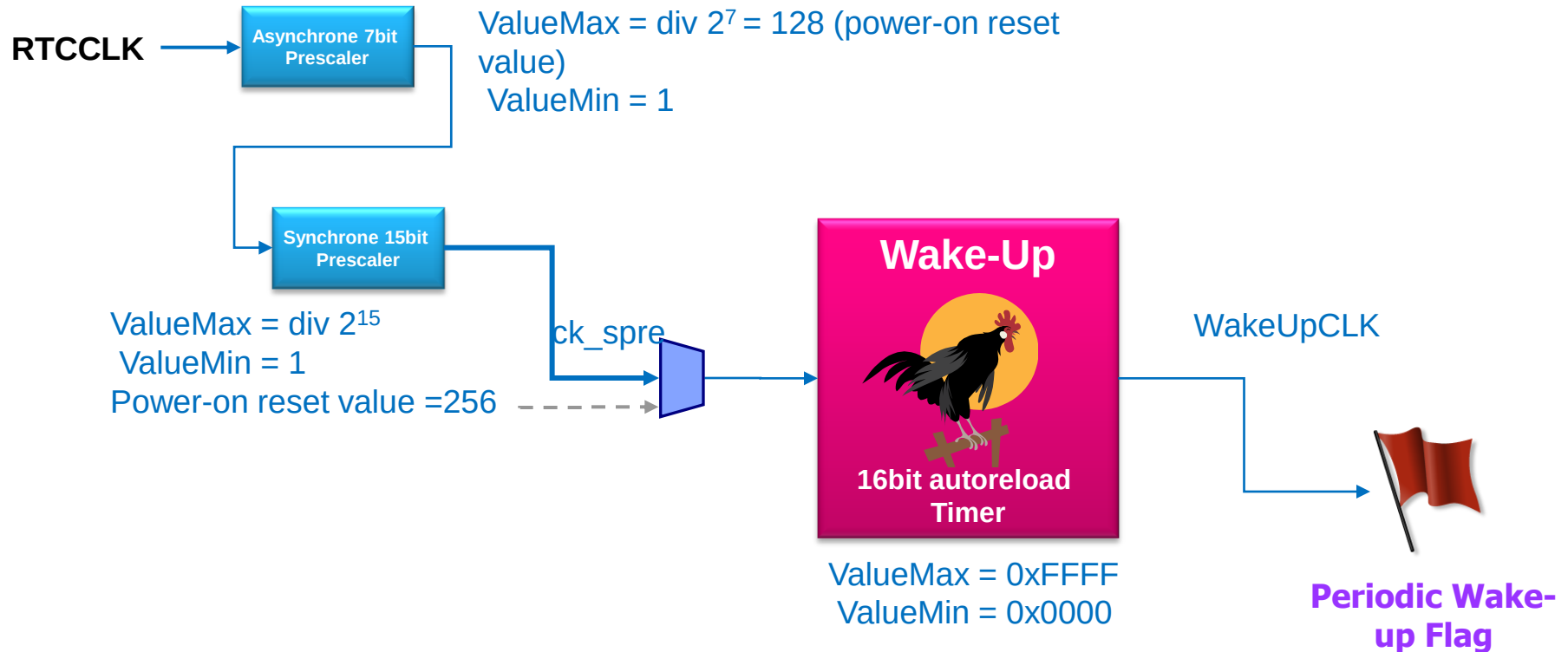
$$\text{WakeUpCLKmin} = \text{RTCCCLK} / (2 \times (0x0001 + 1)) \Rightarrow 122\mu\text{s}$$
$$\text{WakeUpCLKmax} = \text{RTCCCLK} / (16 \times (0xFFFF + 1)) \Rightarrow 32\text{s}$$

$$\text{RTCCCLK} = 32.768\text{KHz}$$
$$\text{Resolution min} = 2 \times \text{RTCCCLK} = 61\mu\text{s}$$

# WakeUp configuration (2/3)

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- Case2 WUCKSEL = 10x



$\text{WakeUpCLKmin} = \text{RTCCLK} / (1 \times (0x0000 + 1))$

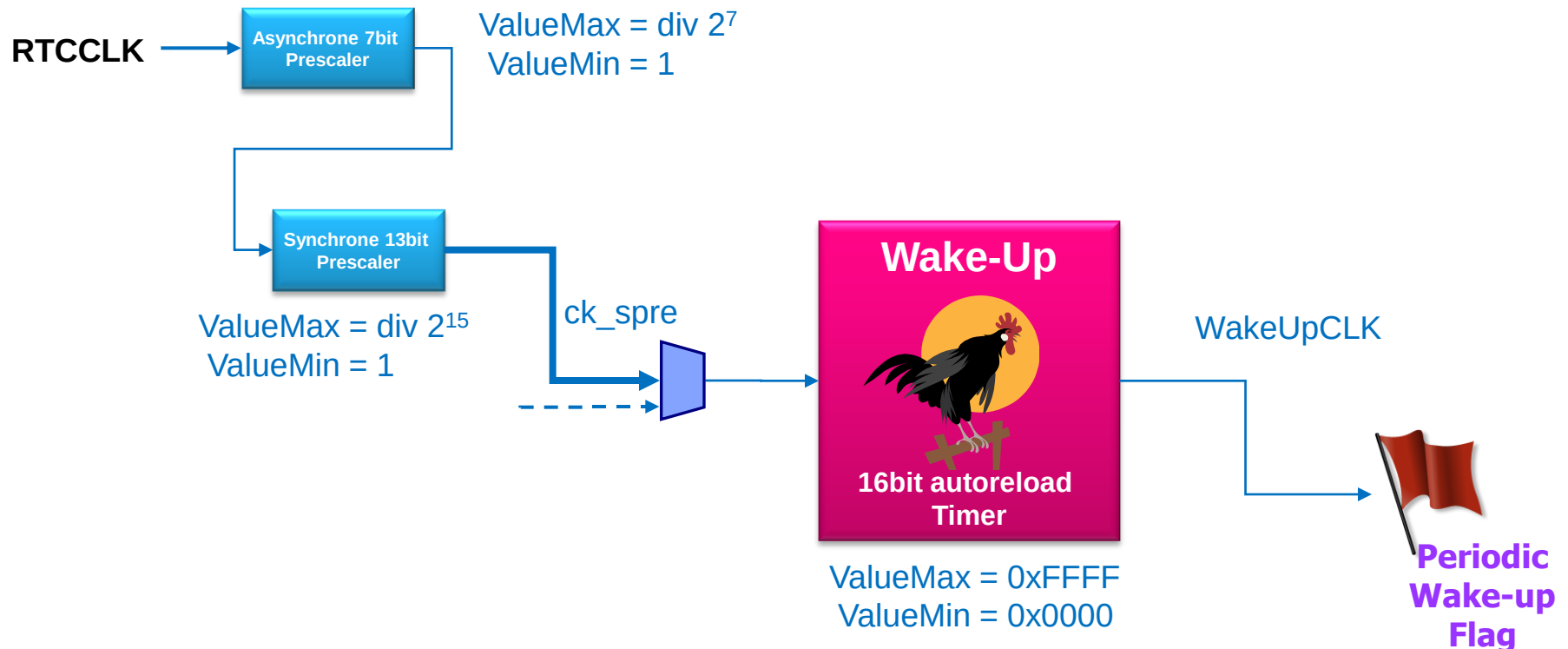
$\text{WakeUpCLKmax} = \text{RTCCLK} / (2^{22} \times (0xFFFF + 1))$

If **ck\_spre** is 1Hz (when used for calendar):  $1s \leq \text{WakeUpCLK} \leq 18.2h$  (1s resolution)

# WakeUp configuration (3/3)

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- Case3 WUCKSEL = 11x



$$\text{WakeUpCLKmin} = \text{RTCCLK} / (1 \times (0x10000 + 1))$$

$$\text{WakeUpCLKmax} = \text{RTCCLK} / (2^{22} \times (0x1FFFF + 1))$$

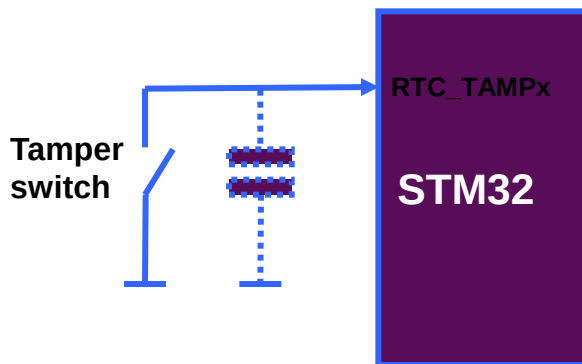
If **ck\_spre** is 1Hz (when used for calendar):  $18.2\text{s} \leq \text{WakeUpCLK} \leq 36.4\text{h}$  (1s resolution)

# Smooth digital calibration

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- Consists in masking/adding N (configurable) 32KHz clock pulses, fairly well distributed in a configurable window.
- A 1Hz output is provided to measure the quartz frequency and the calibration result.

| Calibration window | Accuracy                         | Total range                        |
|--------------------|----------------------------------|------------------------------------|
| 8 s                | <b><math>\pm 1.91</math> ppm</b> | <b>[0 <math>\pm 480</math>ppm]</b> |
| 16s                | <b><math>\pm 0.95</math> ppm</b> | <b>[0 <math>\pm 480</math>ppm]</b> |
| 32s                | <b><math>\pm 0.48</math> ppm</b> | <b>[0 <math>\pm 480</math>ppm]</b> |



Capacitor is optional (filtering can be done by software)

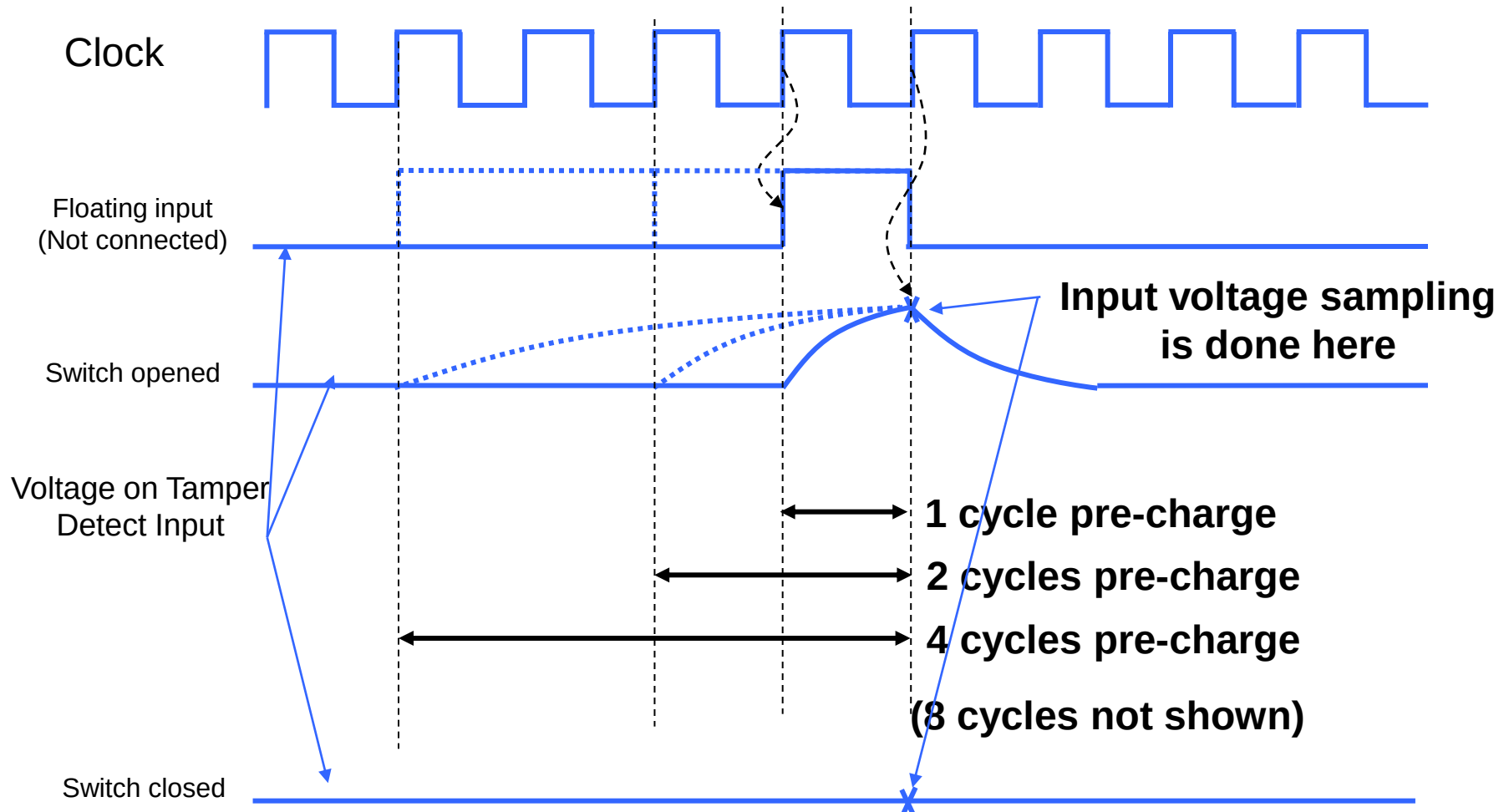
Biasing is done using the I/O's Pull-up resistor

- 3 tamper pins and events
- Configurable active level for each event
- Configurable use of I/Os pull-up resistors
- Configurable pre-charging pulse to support different capacitance values
  - 1, 2, 4 or 8 cycles
- Configurable filter:
  - Sampling rate : 128Hz, 64Hz, 32Hz, 16Hz, 8Hz, 4Hz, 2Hz, 1Hz
  - Number of consecutive identical events before issuing an interrupt to wake-up the MCU : 1, 2, 4, 8

- RTC\_TAMP1 available in VBAT mode.
- Reset of backup registers when tamper event detected
- Tamper event can generate a timestamp event

# Tamper detection - signals

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- What are the different RTC clock sources ?

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- What are the different RTC interrupts ?

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- What is the maximum RTC Sub second (RTC\_SSR) resolution?

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- How many RTC Backup Registers are available?

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Thank you !

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